The sphere and its sections; inscribed and circumscribed solids

Every plane section is a circle — and that single fact settles every sphere problem.

LESSON 41–42 | 2 × 40 MIN | GEOMETRY 11, §19 | IGCSE E13.X

LESSON CLOCK

12' | 24' | 24' | 24' | 20' | 6'

Sphere and ball	12 min
Plane sections	24 min
Tangent planes	24 min
Inscribed and circumscribed	24 min
Practice	20 min
Homework	6 min

LEARNING OBJECTIVES

- Define a sphere and a ball, and name their elements.
- Compute the radius of a plane section from its distance to the centre.
- Recognise great and small circles.
- Solve problems on solids inscribed in or circumscribed about a sphere.

TEXTBOOK

National	pp. 205–216
Cambridge	Core & Extended, pp. 347–354
Workbook	Exercise 19.1

01 Explanation

Sphere and ball

TWO DIFFERENT OBJECTS

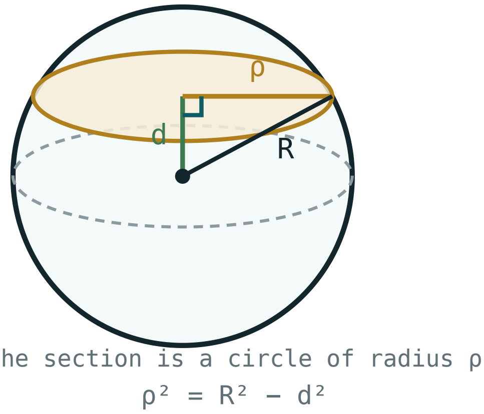
A **sphere** is the **surface** — all points at distance R from the centre. A **ball** is the solid — all points at distance $\le R$. Surface area belongs to the sphere; volume belongs to the ball.

In coordinates (Quarter I) the sphere of centre (a, b, c) and radius R is

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

The sphere has infinitely many planes and axes of symmetry, and one centre — the most symmetric solid there is.

Plane sections



A plane at distance d from the centre cuts a circle of radius ρ, with $\rho^2 = R^2 - d^2$.

$$\rho^2 = R^2 - d^2$$

D	SECTION	NAME
0	a circle of radius R	a **great** circle
$0 < d < R$	a circle of radius $\sqrt{R^2 - d^2}$	a **small** circle
$d = R$	a single point	a tangent plane
$d > R$	nothing	the plane misses the sphere

The great circles are the largest sections, and the shortest path between two points on a sphere runs along one — which is why aircraft routes look curved on a flat map.

Tangent planes

TANGENT PLANE

A plane at distance exactly R from the centre touches the sphere at one point, and is **perpendicular** to the radius drawn to that point.

That perpendicularity is the space version of the circle theorem of Grade 8, and it is proved the same way: any other point of the plane is further from the centre than the foot of the perpendicular, so it lies outside the sphere.

Inscribed and circumscribed

SITUATION	RELATION
sphere inscribed in a cube of edge a	$R = \frac{a}{2}$
sphere circumscribed about a cube of edge a	$R = \frac{a\sqrt{3}}{2}$
sphere inscribed in a cylinder of radius r, height $2r$	$R = r$
sphere circumscribed about a cuboid $a \times b \times c$	$R = \frac{\sqrt{a^2 + b^2 + c^2}}{2}$

THE TWO QUESTIONS TO ASK

Inscribed: the sphere touches the faces, so its diameter is the smallest dimension. **Circumscribed:** the sphere passes through the vertices, so its diameter is the space diagonal.

Archimedes' result. A sphere inscribed in a cylinder of the same radius and height $2r$ has exactly $\frac{2}{3}$ of the cylinder's volume — and exactly $\frac{2}{3}$ of its total surface area. Archimedes had it carved on his tomb.

02 Worked examples

EXAMPLE 1 **A plane is *8* cm from the centre of a sphere of radius *17* cm. Find the radius of the section.**

1. $\rho^2 = 289 - 64 = 225$
2. $\rho = 15$

ANSWER

15 cm

EXAMPLE 2

A sphere is inscribed in a cube of edge 12. Find its radius. Then find the radius of the sphere through the cube's vertices.

1. Inscribed: touches the faces, $2R = 12$.
   $R = 6$
2. Circumscribed: passes through the vertices.
   $2R = 12\sqrt{3}$
3. $R = 6\sqrt{3} \approx 10.4$

ANSWER

6 and $6\sqrt{3}$

EXAMPLE 3

A section of a sphere has radius 12 and is 9 from the centre. Find the sphere's radius and its surface area.

1. $R^2 = 144 + 81 = 225$
2. $R = 15$
3. $S = 4\pi(225) = 900\pi \approx 2827$

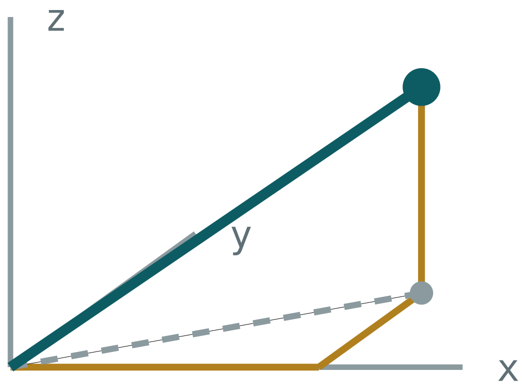
ANSWER

$R = 15$; $900\pi \approx 2830$

03 Interactive model

Cut an orange at different heights and compare the circles.

Cutting a sphere



x **3** | y **2** | z **2** | OM **4.12** | base diagonal **3.61**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Sphere	Sfera	Сфера
Ball	Shar	Шар
Great circle	Katta aylana	Большая окружность
Small circle	Kichik aylana	Малая окружность
Section of a sphere	Sfera kesimi	Сечение сферы
Tangent plane	Urinma tekislik	Касательная плоскость
Inscribed sphere	Ichki chizilgan sfera	Вписанная сфера
Circumscribed sphere	Tashqi chizilgan sfera	Описанная сфера
Diameter	Diametr	Диаметр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Every plane section of a sphere is:

- an ellipse
- a circle
- a parabola
- a point

ρ^2 equals:

- $R^2 + d^2$
- $R^2 - d^2$
- $d^2 - R^2$
- Rd

A great circle has d equal to:

- R
- 0
- $\frac{R}{2}$
- $2R$

A sphere inscribed in a cube of edge a has radius:

- a
- $\frac{a}{2}$
- $\frac{a\sqrt{3}}{2}$
- $a\sqrt{2}$

A sphere through a cube's vertices has radius:

- $\frac{a}{2}$
- $\frac{a\sqrt{2}}{2}$
- $\frac{a\sqrt{3}}{2}$
- a

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. ρ with $R = 5, d = 3$

ANSWER 4

2. ρ with $R = 13, d = 5$

ANSWER 12

3. ρ with $R = 17, d = 8$

ANSWER 15

4. Section at $d = 0$ is called

ANSWER a great circle

5. Section at $d = R$ is

ANSWER a single point

6. Sphere inscribed in a cube of edge 10: R

ANSWER 5

7. Sphere round a cube of edge 10: R

ANSWER $5\sqrt{3}$

Medium · the standard question

7 PROBLEMS

1. Section radius 12 at $d = 9$: the sphere's radius

   ANSWER 15

2. Same: the surface area

   ANSWER 900π

3. Sphere round a cuboid $3 \times 4 \times 12$: its radius

   ANSWER 6.5

4. Sphere inscribed in a cylinder $r = 6$, $h = 12$: R

   ANSWER 6

5. A plane 10 from the centre of a sphere of radius 26: the section radius

   ANSWER 24

6. A section of radius 8 in a sphere of radius 10: its distance

   ANSWER 6

7. A plane 15 from a sphere of radius 12 cuts

   ANSWER nothing

Hard · stretch and reason

7 PROBLEMS

1. A sphere of radius 25; two parallel planes on the same side at 7 and 20. The section radii

   ANSWER 24 and 15

2. A cone of base radius 6 and height 8 inscribed in a sphere: the sphere's radius

   ANSWER 6.25

3. Archimedes: the sphere's volume as a fraction of its cylinder

   ANSWER $\frac{2}{3}$

4. And the surface areas

   ANSWER $\frac{2}{3}$ as well

5. A sphere touches all 12 edges of a cube of edge a: its radius

   ANSWER $\frac{a\sqrt{2}}{2}$

6. Two spheres of radii 5 and 3 with centres 6 apart: the radius of the common circle

   ANSWER ≈ 2.99

7. Prove a tangent plane is perpendicular to the radius at the point of contact

   ANSWER Any other point is further from the centre

07 Homework

Homework — 5 tasks

Draw the right triangle with legs d and ρ and hypotenuse R every time.

1. A plane is 9 cm from the centre of a sphere of radius 41 cm. Find the section radius.
2. A section of radius 20 lies 21 from the centre. Find the sphere's radius and surface area.
3. Find the radius of the sphere inscribed in, and of the sphere circumscribed about, a cube of edge 8.

4. Find the radius of the sphere through the vertices of a $6 \times 8 \times 24$ cuboid.
5. Explain in three sentences the difference between a sphere and a ball.